

CS PhD Seminar Series

September 29th

| 14:30-15:30

| Room 214

Moving Minds, Producing Science: A Computational Study of Researcher Mobility and Regional Scientific Productivity

How is researcher mobility associated with scientific productivity across regions? Building on the work in Scientific Mobility group at the Max Planck Institute for Demographic Research, this talk examines international and internal researcher mobility across subnational regions worldwide using longitudinal publication and affiliation data. Using the Scholarly Migration Database 2.0 as our main source, we compared differences across regions with changes within the same region over time, combining descriptive analysis, panel models, and quasi-experimental approaches. From a computational perspective, the study highlights challenges that also arise in machine learning: constructing meaningful variables from large behavioral datasets, modeling temporal and geographic dependence, addressing selection bias, and separating prediction from causal interpretation. The results show that highly mobile regions can be more productive, while increases in mobility within a region do not consistently correspond to productivity gains.



Speaker: **Marco Monteverde**

Marco Monteverde is a Ph.D. student at DIBRIS under the supervision of Prof. Annalisa Barla. Reflecting his highly interdisciplinary background, his research lies at the intersection of computer science, machine learning, and political science. This summer, he spent three months as a visiting researcher at the Max Planck Institute for Demographic Research in Rostock, Germany, where he worked on a research project in Science of Science and demography with an international, interdisciplinary research team.

Learned Regularization for Image Deconvolution in Fluorescence Microscopy

In imaging, the visual data we capture is a degraded version corrupted by blur and noise. Reconstructing the true underlying signal from these imperfect measurements defines an imaging inverse problem—a fundamental challenge in modern optical microscopy, where light diffraction causes blurring and photon starvation introduces heavy noise. To overcome these limits, Image Scanning Microscopy (ISM) improves resolution by replacing the confocal pinhole with a small detector array: a 5×5 SPAD array which produces 25 images, each formed through its own shifted point spread function. We formulate the reconstruction as a Poisson inverse problem and solve it in a maximum a posteriori framework. The data fidelity term is a multi-detector Kullback–Leibler divergence and the prior is supplied by explicit regularization, either a smoothed total variation functional or an ℓ_1 penalty, promoting piecewise-smooth or sparse solutions. Minimization is carried out by projected gradient descent. Because the same iteration structure accommodates implicit priors, the scheme extends naturally to Plug-and-Play variants, in which the regularization step is replaced by a pre-trained image denoiser. We report results with DRUNet, used both as released for natural images and after fine-tuning on simulated microscopy data, and compare them against the handcrafted regularizers on simulated tubulin phantoms and on experimental acquisitions.

Speaker: **Sofia Agostoni**

Sofia Agostini is a PhD student at MaLGA center in Genoa, affiliated to Genoa University. She has been working on Inverse Problems with applications in Microscopy, developing optimisation-driven learning methods. She focuses on application of the Image Scanning Microscope, with PnP methods and Sampling strategies for uncertainty quantification.

